

Eklavya Sharma

Curriculum Vitae

✉ Email: eklavya2@illinois.edu, ekurn@gmail.com
🌐 Personal website: <https://sharmaeklavya2.github.io>
📞 Phone: +1 217-377-5792
in sharmaeklavya2 [↗](#)  sharmaeklavya2 [↗](#)

Research Interests

Economics and computation, Game theory, Social choice theory, Approximation algorithms

Education

Aug 2021 – present **PhD**, *Department of Industrial & Enterprise Systems Engineering (ISE), University of Illinois at Urbana-Champaign (UIUC), IL, USA*
Research on fair resource allocation. Advised by Prof. Jugal Garg [↗](#).
July 2019 – **M.Tech. (Research)**, *Computer Science and Automation (CSA), Indian Institute of Science (IISc), Bangalore, GPA: 9.7 / 10.0*
Did research on approximation algorithms for variants of bin packing and knapsack. Advised by Prof. Arindam Khan [↗](#).
Aug 2014 – **B.E. (Hons) Computer Science**, *Birla Institute of Technology and Science (BITS), Pilani, India, GPA: 9.14 / 10.00*
June 2018

Publications

- Exploring relations among fairness notions in discrete fair division [↗](#), with Jugal Garg, on ArXiv.
- Improving approximation guarantees for maximin share [↗](#) [↗](#), with Hannaneh Akrami, Jugal Garg, and Setareh Taki, in EC 2024.
- New fairness concepts for allocating indivisible items [↗](#) [↗](#), with Ioannis Caragiannis, Jugal Garg, Nidhi Rathi, and Giovanna Varricchio, in IJCAI 2023.
- Simplification and improvement of MMS approximation [↗](#) [↗](#), with Hannaneh Akrami, Jugal Garg, and Setareh Taki, in IJCAI 2023.
- Two-player matrix games repeated until collision [↗](#), with Aniket Murhekar, in FSTTCS 2023.
- EF1 for mixed manna with unequal entitlements [↗](#), with Jugal Garg, ArXiv.
- Best-of-both-worlds fairness of the envy-cycle-elimination algorithm [↗](#), with Jugal Garg, on ArXiv.

- Approximation algorithms for multidimensional packing[↗], with Arindam Khan and K.V.N. Sreenivas, in FSTTCS 2022.
- Tight approximation algorithms for geometric bin packing with skewed items^{↗↗↗}, with Arindam Khan, in Algorithmica and APPROX 2021.
- Harmonic algorithms for packing d -dimensional cuboids into bins^{↗↗}, in FSTTCS 2021.
- Automating the search for small hard examples to approximation algorithms[↗], on ArXiv.
- An approximation algorithm for covering linear programs and its application to bin-packing[↗], on ArXiv.
- Mitigating DNS amplification attacks using a set of geographically distributed SDN routers[↗], with Vishal Gupta, in ICACCI 2018.

Fellowships and Achievements

- Aug 2024 **Mavis Future Faculty Fellowship, UIUC** [↗]
A program that trains fellows on various aspects of an academic career through workshops, seminars, and activities.
- April 2024 **William A. Chittenden II Award, UIUC** [↗]
Given to an outstanding graduate student at ISE.
- July 2023 **Dr. MNS Swamy Medal for Best MTech (Research) Thesis, IISc** [↗]
- April 2023 **Sharp Outstanding Graduate Student Award, UIUC** [↗]
- Aug 2021 – July 2022 **Samuel Brainin Engineering Fellowship, UIUC**
- March 2018 **Graduate Aptitude Test in Engineering (GATE)**
Secured all-India rank 86 (out of approximately 100,000 candidates) in the ‘Computer Science and IT’ test.
- Aug 2014 – Dec 2015 **BITS Pilani Merit Scholarship**
Scored GPA among top 2% of students in three semesters.
- ACM-ICPC** [↗]
- ACM-ICPC is an international annual multi-tiered programming contest for college students. Around 3000 teams (of 3 students each) participate in the Indian online qualifying round each year. Top few teams qualify for on-site regional contests in India.
- Dec 2017 Ranked 29 out of 250 teams in the Amritapuri regional contest.
- Dec 2016 Ranked 66 out of 450 teams in the Amritapuri regional contest.
- Dec 2016 Ranked 30 out of 70 teams in the Kharagpur regional contest.
- Dec 2015 Ranked 88 out of 250 teams in the Amritapuri regional contest.

Invited Talks

- 27 Oct 2023 **Fair allocation of indivisible items**
Capital Area Theory Seminar, University of Maryland, College Park

22 Dec 2022 **Existence and computation of epistemic EFX allocations** 
Indian Institute of Science, Bangalore

Professional Service

Program committee member for EC'25, AAMAS'25, AAMAS'24.

Conference subreviewer for EC'24, ICALP'24, EC'23, ESA'23, IJCAI'23, FCT'23, STOC'22, SAGT'22, MFCS'21.

Journal reviewer for Optimization Letters (2024).

Research Projects

June 2022 – **Algorithms for Fair Division of Indivisible Items**

Present *Topics:* fair division.

Supervisor: Prof. Jugal Garg , ISE, UIUC.

Jan 2020 – **Approximation Algorithms for Geometric Packing Problems** 

July 2021 *Topics:* approximation algorithms, bin packing.

Supervisor: Prof. Arindam Khan , CSA, IISc Bangalore.

Sept 2017 – **Mitigating DNS-related DoS attacks using SDN** 

Dec 2017 *Topics:* computer networks, network security, SDN.

Supervisor: Prof. Vishal Gupta, BITS Pilani.

Nov 2017 – **CT-means clustering algorithm** 

Jan 2018 *Topics:* machine learning, algorithms, optimization.

Supervisor: Prof. Surekha Bhanot, BITS Pilani.

Invented a clustering algorithm that is a fast approximation to C-means fuzzy clustering. Mathematically proved its convergence and approximation guarantees. Implemented  the algorithm and benchmarked its performance. It was not significantly faster in practice and its applicability was limited.

Work Experience

Teaching Assistant, *IE 300: Analysis of Data (Fall'22, Spring'24, Spring'25)*, UIUC

Spring 2023 **Teaching Assistant**, *IE 310: Deterministic models in optimization*, UIUC

Fall 2020 **Teaching Assistant**, *Design and Analysis of Algorithms*, IISc Bangalore

Aug 2018 – **Software Engineer**, *media.net*, Bangalore, India

July 2019 *Topics:* machine learning, large-scale systems.

media.net is an advertisement-technology company. I worked on improving their machine-learning-based algorithm for bidding in real-time ad auctions.

Jan 2018 – **Intern** , *American Express*, Gurgaon, India

June 2018 *Topics:* neural networks, machine learning, big data.

Trained a neural network to predict credit-card defaulting. The input format was unconventional, so a custom architecture was devised. Its performance was at par with the production model, which was tuned over many years.

May 2017 – **Intern**, *Directi*, Mumbai, India

July 2017 *Topics*: machine learning.

Made Directi's news article classification algorithm recognize more categories.

May 2016 – **Google Summer of Code (GSoC) Student** ↗, *Zulip*

Aug 2016 *Topics*: software development.

Zulip is an open-source group chat application. 3 students were selected from over 100 applicants to work on Zulip as part of the GSoC program.

- Added type annotations to Zulip's python code (around 50,000 lines) so that it could be statically type-checked using a tool called mypy. This improved developer productivity and made Zulip the first major open source project to be 100% statically typed with PEP-484 annotations.
- Switched from an apt repository to using virtualenvs in production. This simplified dependency management and testing deployment workflow.
- Migrated Zulip's python code from Python 2 to Python 3. Apart from a lot of ad-hoc bug-fixing, this involved:
 - Writing scripts which used static code analyzers to find Python 3 bugs.
 - Migrating to python3-compliant dependencies. This required some sections of code to be entirely rewritten.
 - Standardizing the way Zulip uses different kinds of strings (text and byte strings).
- A more detailed description of my work:
<https://gist.github.com/sharmaeklavya2/57c2420865f17fc9b58a78033de61422>.

Selected Coursework

UIUC:

- (CS 580) Algorithmic Game Theory: grade A
- (CS 598 TH1) Recent Advances in Theoretical CS: grade A+
- (CS 473) Algorithms: grade A+
- (IE 511) Integer Programming: grade A
- (IE 519) Combinatorial Optimization: grade A
- (IE 410) Advanced Stochastic Processes and Applications: grade A+
- (IE 411) Optimization of Large Systems: grade A+

IISc Bangalore:

- Approximation Algorithms: grade A+, rank 1
- Design and Analysis of Algorithms: grade A+, rank 1
- Computational Methods of Optimization: grade A+, rank 1
- Cryptography: grade A

BITS Pilani: Advanced Algorithms and Complexity, Discrete Structures in Computer Science, Design and Analysis of Algorithms, Cryptography, Graphs and Networks, Theory of Computation, Data Structures and Algorithms, Logic in Computer Science, Machine Learning.

Computer Skills

L^AT_EX, Python, HTML, CSS, JavaScript, C/C++, Java, SQL, Bash.

Student Societies

BITS-ACM, *BITS Pilani ACM Student Chapter*

- Problem setter for 6 programming contests organized by BITS-ACM.
- Created backends for web applications used in online quizzing events.
- Conducted intra-BITS-ACM workshops on ‘Competitive Programming’ and ‘Linux and CLI’.

References

Jugal Garg

Assistant Professor, ISE, UIUC

✉ jugal@illinois.edu

🌐 <https://jugal.ise.illinois.edu/>

Arindam Khan

Assistant Professor, CSA, IISc Bangalore

✉ arindamkhan@iisc.ac.in

🌐 <https://www.csa.iisc.ac.in/~arindamkhan/>